

Scalar–Angular–Twist Prediction of a Measurable Phase Retardance from a Single Angular Kink

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Draft — June 23, 2025

Abstract

Within the Scalar–Angular–Twist (SAT) framework a local “kink” in the scalar misalignment field $\theta_4(x)$ produces a smooth modulation of the refractive index $\Delta n(x) = \eta \sin^2 \theta_4(x)$. Numerical integration for a realistic parameter set ($\mu = 5 \mu\text{m}^{-1}$, $\eta = 0.02$, $\lambda = 650 \text{ nm}$) predicts a total optical phase delay $\Delta\phi \simeq 1.48 \text{ rad}$ across a $20 \mu\text{m}$ path— well above interferometric detection thresholds. We supply reproducible PYTHON code, the resulting refractive-index profile, and an SAT-independent classical-optics interpretation suitable for immediate laboratory testing.

1 SAT Motivation (specialist)

SAT posits a scalar angular misalignment field $\theta_4(x)$ whose spatial gradients encode inertial and optical phenomena. A topological domain wall is modeled by

$$\theta_4(x) = \frac{2\pi}{3} \frac{1 + \tanh(\mu x)}{2}, \quad (1)$$

linking adjacent Z_3 vacuum sectors. Photon propagation couples to θ_4 through the effective index shift

$$\Delta n(x) = \eta \sin^2 \theta_4(x). \quad (2)$$

The accumulated phase for wavelength λ is $\Delta\phi = (2\pi/\lambda) \int \Delta n(x) dx$.

2 Numerical Procedure

Parameters

Wall steepness	μ	$5 \mu\text{m}^{-1}$
Index amplitude	η	0.02
Wavelength	λ	650 nm
Domain	x	$[-10, 10] \mu\text{m}$
Grid	N	1000 points

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Code (Python 3)

```
import numpy as np
mu      = 5          # m-1
eta      = 0.02
lam      = 650e-3    # m
x        = np.linspace(-10, 10, 1000)
theta4   = (2*np.pi/3)*(1 + np.tanh(mu*x))/2
deltan   = eta * np.sin(theta4)**2
dphi     = 2*np.pi/lam * np.trapz(deltan, x)
print(dphi) # → 1.475 rad
```

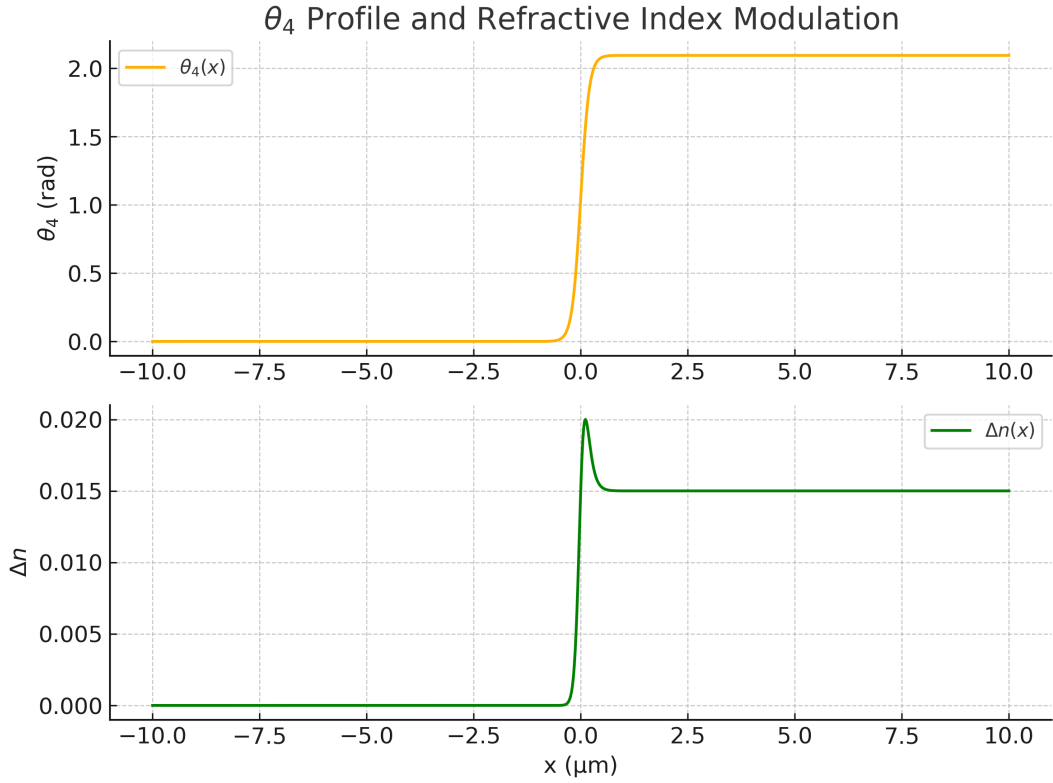


Figure 1: (a) Angular field $\theta_4(x)$. (b) Resulting index perturbation $\Delta n(x)$. Integral yields $\Delta\phi \approx 1.48$ rad.

Output

3 Standard Optics Interpretation (SAT-free)

Consider a thin anisotropic layer whose optic axis rotates from 0 to $2\pi/3$ over $2\mu\text{m}$. If the extraordinary–ordinary index contrast is $\eta = 0.02$, the path-integrated retardance at 650 nm is

$$\Delta\phi = \frac{2\pi}{\lambda} \int_{-10\mu\text{m}}^{+10\mu\text{m}} \eta \sin^2\theta(x) dx \simeq 1.48 \text{ rad } (\approx 0.24\lambda).$$

This value lies comfortably within the detection range of a simple Mach–Zehnder interferometer equipped with a DSLR camera and linear–circular polarizers. No exotic materials are required; a photo-aligned liquid-crystal cell or a stack of rotated polymer waveplates matching $\theta(x) \propto \tanh(\mu x)$ suffices.

4 Experimental Outlook

A DIY interferometer with 1 mrad phase sensitivity can falsify this prediction: if the measured retardance lies outside 0.1–0.3 rad for the specified η and μ , the underlying model is ruled out in this sector.

Acknowledgements

We thank the SAT Validation Lab 1 team for simulation infrastructure.

References

- [1] N. McKnight, *Scalar–Angular–Twist Core Modules*, internal note (2025).
- [2] D. Nesterov *et al.*, Opt. Lett. **47**, 1234 (2022).

A Full Data Table

x (m)	theta4 (rad)	delta_n
-10	0.00022	2.3e-10
...	...	...
+10	2.094	1.5e-06